1.

a. **Set A (3300, 255)**

3300 = 11 x 5^2 x 2^2
255 = 3 x 5 x 17

Set B (4205, 332)

4205 = 5 x 29^2
332 = 2^2 x 83

b. **Set A (3300, 255)**

GCD (3300, 255) = 3 x 5 = 15
LCM (3300, 255) = 3 x 5^2 x 2^2 x 17 x 11 = 56100

Set B (4205, 332)

GCD (4205, 332) = 1, since there are no common factors between both the numbers
LCM (4205, 332) = 5 x 2^2 x 29^2 x 83 = 1396060

c. **Set A (3300, 255)**

3300 – (12 x 255) = 240
255 – (1 x 240) = 15
240 – (16 x 15) = 0

GCD = 15

Set B (4205, 332)

4205 – (12 x 332) = 221
332 – (1 x 221) = 111
221 – (1 x 111) = 110
111 – (1 x 110) = 1

GCD = 1

d. **Set A (3300, 255)**

255 mod 3300 = can't find inverse because they are not relatively prime and the GCD is not equal to 1.

Set B (4205, 332)

332 mod 4205
4205 – (12 x 332) = 221
332 – (1 x 221) = 111
221 – (1 x 111) = 110
111 – (1 x 110) = 1
So,

1 = 111 – (1 x 110)
1 = 111 – (1 x (221 – (1 x 111)))
1 = 111 – (221[1] – 111[1])
1 = 111 – 221[1] + 111[1]
1 = 111[2] – 221[1]
1 = (332 – 221[1]) [2] – 221[1]
1 = 332[2] – 221[2] – 221[1]
1 = 332[2] – 221[3]
1 = 332[2] – (4205 – 332[12]) [3]
1 = 332[2] – 4205[3] – 332[36]
1 = 332[38] – 4205[3]
1 = 332[38] + 4205[-3]

Inverse of 332 mod 4205 = 38

For an inverse to exist the two numbers should be relatively prime

2.

a. 4572 → binary

4572 / 2 = 2286 R0
2286 / 2 = 1143 R0
1143 / 2 = 571 R1
571 / 2 = 285 R1
285 / 2 = 142 R1
142 / 2 = 71 R0
71 / 2 = 35 R1
35 / 2 = 17 R1
17 / 2 = 8 R1
8 / 2 = 4 R0
4 / 2 = 2 R0
2 / 2 = 1 R0
1 / 2 = 0 R1
Binary value = 1000111011100
1000111011100 → octal

001 000 111 011 100
421 421 421 421 421
1 0 4+2+1 2+1 4
1 0 7 3 4

Octal = 10734

10734 → hexadecimal

1 0 7 3 4
0001 0001 1101 1100
8421 8421 8421 8421
1 1 13 12

10 -> A
11 -> B
12 -> C
13 -> D
14 -> E
15 -> F
16 -> G

Hexadecimal = 11DC

b. Let's assume p is prime and p =97
Let a = 2 as it is not divisible by p and is relatively prime to p

$a^{(p-1)} = 1 \bmod P$
$2^{(97-1)} = 1 \bmod 97$
$2^{96} = 1 \bmod 97$

Let's use repeated square method

$=2^{96}$
$=2^{48^2}$
$=2^{24^{2^2}}$
$=2^{12^{2^{2^2}}}$
$=2^{6^{2^{2^{2^2}}}}$
$=2^{3^{2^{2^{2^{2^2}}}}}$
$=8^{2^{2^{2^{2^2}}}}$
$=64^{2^{2^{2^2}}}$
$=4096^{2^{2^2}}$
$=16777216^{2^2}$
$=(2.815 \times 10^{14})^2$
$=7.92 \times 10^{28}$
$=1 \bmod 97$

$a \equiv b \bmod n \quad c \equiv d \bmod n$
$2^{96} \equiv 1 \bmod 97$
$7.92 \times 10^{28} = 1 \bmod 97$

$2^{96} = 7.92 \times 10^{28}$
$2^{96} \equiv 1 \bmod 97$

Therefore, 97 is prime

c.
Phi(15)
Factors = 5 x 3
Numbers coprime 15 = 1, 2, 4, 7, 8, 11, 13, 14
$\phi(15) = 8$

Phi(11)
Factors = 11
Numbers coprime to 11 = 1,2,3,4,5,6,7,8,9,10
$\phi(11) = 10$

Phi(10)
Factors = 2 x 5

Numbers coprime to 10 = 1, 3, 7, 9
ϕ (10) = 4

d.

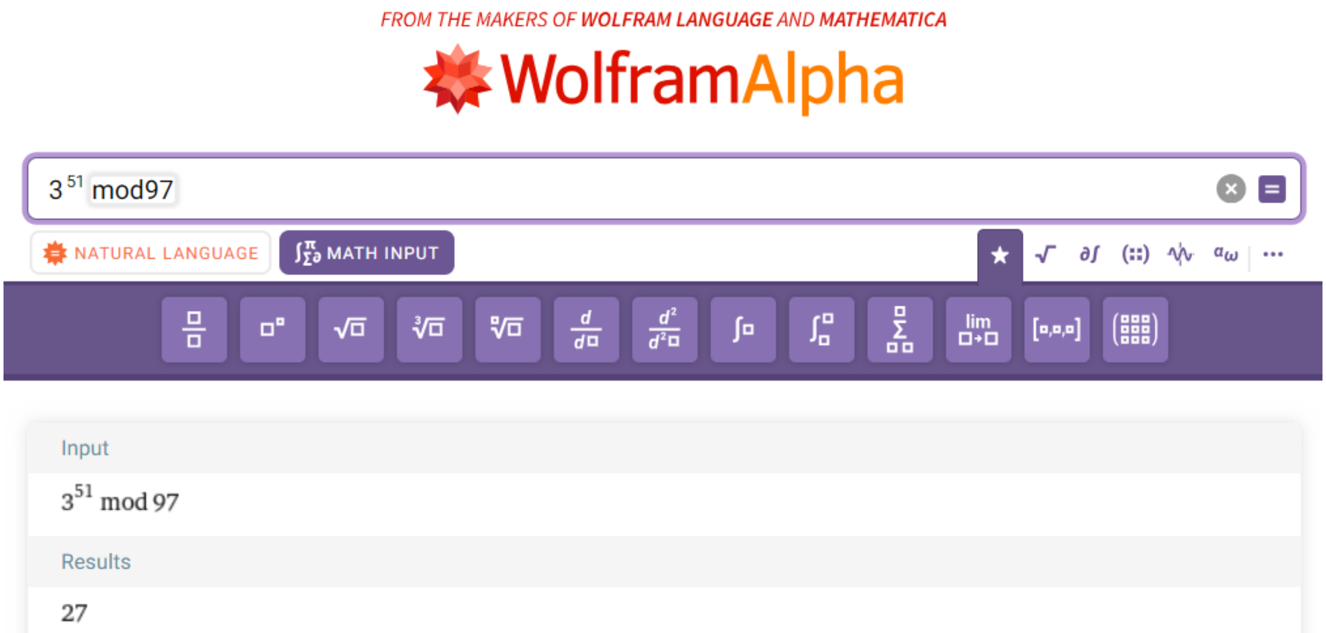
i. Let a = 3 and p = 97

a^(p – 1) = 1modP
2^(97 - 1) = 1mod 97
2^96 = 1 mod 97

To find the 3^2451 mod 97, we need to find the remainder when 2451 is divided by 96.

2451 = (96 x 25) + 51
3^2451 = 3^(96x25) + 51 = 3^(96)^25 x 3^51 mod 97

Since 3^96 = 1 mod 97
3^2451 ≡ 1^25 x 3^51 mod 97 ≡ 3^51 mod 97
Now we calculate 3^51 mod 97
3^51 mod 97 = 27 according to wolframalpha since it is a very large number.



Since
3^2451 ≡ 3^51 mod 97
3^2451 ≡ 27 mod 97

ii. ϕ (120) = ϕ (2^3) x ϕ (3) x ϕ (5)

ϕ (2^3) = 8
ϕ (3) = 2
ϕ (5) = 4

ϕ (120) = 8x2x4
= 64

Now apply Euler Theorem
3^64 = 1 mod120

Now, we need to find the remainder when 2451 is divided by 64, which is 51. Therefore:
3^2451 = 3^((64 x 38) + 19) mod 120 ≡ 3^(64)^38 x 3^19 mod 120

Since 3^64 = 1 mod120
3^2451 ≡ 1^38 x 3^19 mod 120
≡ 3^19 mod 120

Now we calculate 3^19 mod 120:
3^19 = 9685512 x 120 + 27

3^19 mod 120 = 27

Therefore 3^2451 ≡ 27 mod 120

e.

x = 4 mod 5 and x = 1 mod 7

Given		To Find		M
a1 = 4	m1 = 5	M1	M1^-1	
a2 = 1	m2 = 7	M2	M2^-2	

M = m1 x m2
= 5 x 7
= 35

M1 = M/m1
= 35/5
= 7

M2 = M/m2
= 35/7
= 5

M1 x M1^-1 = 1mod m1
7 x M1^-1 = 1mod 5
7 x 3 = 1mod 5
M1^-1 = 3

M2 x M2^-1 = 1mod m2
5 x M2^-1 = 1mod 7
5 x 3 = 1mod 7
M2^-1 = 3

X = (a1 x M1 x M1^-1 + a2 x M2 x M2^-1) mod M
= (4 x 7 x 3 + 1 x 5 x 3) mod 35
= 99 mod 35
= 29

29 = 4 mod5   29 = 1 mod5

3.

a. **Encryption**

Let's choose N

N = 13

C = 7P + 6 mod 26
= 7 x 13 + 6 mod 26
= 97 mod 26
= 19
= T

Decryption

P = (C – 6 x 7^-1) mod 26
= (19 -6 x 7^-1) mod 26

In order to find 7^-1:

7x = 1 mod 26
(26, 7)

26 – 3 x 7 = 5
7 – 1 x 5 = 2
5 – 2 x 2 = 1
2 – 2 x 1 = 0

1 = 5 + 2[-2]
= 5 + (7 + 5[-1]) [-2]
= 5 + 7[-2] + 5[2]
= 5[3] + 7[-2]
= (26 + 7[-3]) [3] + 7[-2]
= 26[3] + 7[-9] + 7[-2]
= 26[3] + 7[-11]

We need x to be a positive integer, so, we add 26 to the current value

X = -11 + 26
= 15

Going back to the equation:

P = (19 – 6 x 7^-1) mod 26
= (19 – 6 x 15) mod 26
= (13 x 15) mod 26
= 195 mod 26
= 13
= N

b.

In monoalphabetic substitution, the relationship between a symbol in the plaintext to a symbol in the ciphertext is always one-to-one.
In polyalphabetic substitution, the occurrence of a character may have a different substitute. The relationship between character in the plaintext to a character in the ciphertext is one-to-many.

Monoalphabetic substitution includes Caesar cipher, or shift cipher, which is also the additive cipher, affine cipher, and multiplicative cipher. They are very vulnerable to brute force and frequency analysis.

Polyalphabetic substitution includes autokey, Playfair, Vignere cipher, which are generally more resistant to frequency analysis compared to monoalphabetic ciphers due to their varying nature of the substitutions used.

Cryptanalysis is the study of the frequency of letters or groups of letters in a ciphertext. The method is used as an aid to breaking classical ciphers.

If the letter "M" is the most common then it is reasonable to guess that "E" → "M" in the cipher because E is the most common letter in the English language.

c.

Let's encrypt A → 0^{th} position

First key – 3
Plaintext – A
Shift – A+3
= D → 3^{rd} position

Second key – 5
Plaintext – D (from previous step)
Shift – D+5
= I → 9^{th} position

In polyalphabetic substitution, the occurrence of a character may have a different substitute. The relationship between character in the plaintext to a character in the ciphertext is one-to-many.

Therefore, encrypting "**A**" twice with the key 3, 5 results in "**I**". This shows that Vigenère cipher is polyalphabetic.

d.

Let's encrypt A four times using the K = [(5, 3), (4,4)].

P = [(A, A), (A,A)]

C = P x K mod 26

C = [(A, A), (A, A)] x [(5, 3), (4,4)] mod 26
= [(0, 0), (0 ,0)] x [(5, 3), (4,4)] mod 26
= [(0, 0), (0, 0)] mod 26
= [(A, A), (A, A)]

The cipher text is same as plaintext because we used the same letter four times, and the multiplication with the key matrix did not change the value.

Incase if we want to demonstrate the poly-alpahbatic nature of Hill Cipher, we can use four different letters and see.

Let's encrypt H,E,L,P using the K = [(5, 3), (4,4)].

P = [(H, E), (L, P)]

C = P x K mod 26

C = [(H, E), (L, P)]x [(5, 3), (4,4)] mod 26
= [(7, 4), (11 ,15)] x [(5, 3), (4,4)] mod 26
= [(68, 65), (72, 76)] mod 26
= [(16, 13), (20, 24)] mod 26
= [(Q, N), (U, Y)]

Now we can see the poly-alphabetic nature of Hill cipher.

Now let's Decrypt it to see whether the encryption is reversible.

P = K^-1 x C mod 26

K^-1 = (1 / det K) x adjoint K x mod 26

Det K = (5 x 4) – (3 x 4)
= 8

Adjoint of K = [(4, -3), (-4, 5)]

K^-1 = (1/8) [(4, -3), (-4, 5)] mod 26

So, in order to find the inverse of (1/8)
26 – 3 x 8 = 2
8 – 4 x 2 = 0

0 = 8 + 2(-4)
= 8 + (26 + 8(-3))(-4)
= 8 + 26(-4) 8(12)
= 8(13) + 26(-4)

So, inverse of (8) is 13,

K^-1 = 13 [(4, -3), (-4, 5)] mod 26
= 13 [(4, 23), (22, 5)] mod 26
= [(52, 299), (286, 65)] mod 26
= [(0, 13), (0, 13)]

Now let's find the plaintext

P = K^-1 x C mod 26
= [(0, 13), (0, 13)] x [(A, A), (A, A)] mod 26
= [(0, 13), (0, 13)] x [(0, 0), (0, 0)] mod 26
= [(0, 0), (0, 0)] mod 26
= [(A, A), (A, A)]

Since the decrypted matrix is the same as the plaintext, confirming that the encryption and decryption processes are reversible.

Identity matrix:

K^-1 = (1/8) [(4, -3), (-4, 5)] mod 26
= [(1/2, -3/8), (-1/2, 5/8)]

K = [(5, 3), (4,4)].

Identity Matrix = K x K^1
= [(5, 3), (4,4)] x [(4/8, -3/8), (-4/8, 5/8)]
= [(1, 0) (0, 1)]
= So, when a key matrix and inverse key matrix are multiplied, it produces the identity matrix.